MICHAEL C. GAUTHIER

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EDUCATION

Northeastern University, College of Engineering Boston, MA
Candidate for Bachelor of Science in Mechanical Engineering, Minor in Entrepreneurial Startups | Dean's List *May 2027*
Relevant Coursework: Thermal Systems Analysis & Design, Measurement & Analysis with Thermal Applications, Mechanical Engineering Computation & Design, Dynamics, Thermodynamics, Fluid Dynamics, Mechanics of Materials
Semester Abroad: Saint Louis University, Madrid, Spain *Fall 2023*
Westfield High School | *Summa Cum Laude* | *Dean's List* *June 2023*

TECHNICAL SKILLS

- **CAD/Design:** AutoCAD 2D/3D, SolidWorks, ArcGIS, Fusion 360, GD&T, plan set production, CAD Standards, Details, Profiles
- **Programming/Analysis:** MATLAB, Python, C++, Java, Arduino, Abaqus, FEA, CST, automation/data analysis, Microsoft Office
- **Lab Work:** Instron tension/compression/bending/torsion, hardness/impact testing, stress-strain & yield/failure analysis
- **Equipment:** Total Station, Trimble GPS/GNSS, 3D printing, laser cutting, X-ray diffraction, RTD/TC, power tools, woodworks

WORK EXPERIENCE

Town of Wellesley, Engineering Division Wellesley, MA
Construction Engineering Co-op *Jul 2025 – Present*

- Designing & implementing an ArcGIS-based inspection system to replace paper forms, cutting inspection time by 50%
- Automating drafting by writing Python scripts to convert 100+ hand-drawn sketches into CAD linework
- Operating Trimble GPS & total station to capture survey points (300+) for plan development
- Drafting AutoCAD drawings (20+), including existing conditions, proposals, and final plan sets

Brandy Melville Boston, MA
Stock Associate *Sep 2024 – Jul 2025*

- Managed all inventory by processing 25+ shipments daily, maintaining organization, and communicating with corporate
- Remodeled backstock by moving storage zones & building shelving to improve workflow efficiency

CA Light Designs Westfield, NJ
Small Business Founder & Owner *Mar 2021 – Aug 2024*

- Manufactured LED products using CAD layouts, engraving machines, and customer-specific design requirements
- Optimized an engraving machine to improve precision and reduce production time by 75%
- Doubled order volume within 1 month through improved production workflow, product quality, and customer service

ENGINEERING PROJECTS

Capstone Project | *Automated Microscope Platform for Microfluidic Imaging* *May 2026 - Present*

- Designing a motorized chip-positioning platform with one-pixel accuracy to hold, move, and align microfluidic samples for accurate high-speed microscope imaging

NU Give a Hand | *Prosthetic Design Team* Jan 2025 – Present

- Modeling prosthetic hand designs in SolidWorks, 3D-printing prototypes, and collaborating to refine designs

NU Electric Racing | *Tire Optimization Team* *Sep 2024 - Present*

- Evaluating tire materials to reduce rotational inertia and improve acceleration/efficiency, using MATLAB to analyze data

Clean Energy Exhibit for Boston Middle School Exposition *Jan 2024 – Apr 2024*

- 3D-printed a SolidWorks wind turbine and laser-cut a CAD house model with LEDs to visualize energy generation
- Programmed a MATLAB Arduino game with interactive lighting that ran glitch-free under repeated child use

VOLUNTEER WORK

Coding with a Cause Club | Co-*President & Co-Founder* *Sep 2022 – Jun 2023*

- Led monthly outreaches to teach kids web design & coding, expanded to 2 schools, and each student built a website

Dana-Farber Cancer Institute | *Fundraising Event Organizer* *Jul 2025 – Oct 2025*

- Planned fundraiser yoga events (instructor sourcing, payment workflow), raising ~$1k for cancer research

INTERESTS

Running (Marathons & Triathlons), Hiking (National Parks), Piano, Cars, Skiing (Ski Club), ASME